

Two 20 nm planar featured Xilinx Kintex Ultrascale (KU) FPGAs distribute clocks and configuration received from the GBTx to the MB, while reading slow integrator data and fast digitized data from two different gains of the PMTs. The xADCs of the KU FPGAs sample temperature, voltage stability and currents. An over-current protection circuit triggers a power cycle in the case of the presence of high currents. Each KU FPGA sends data off-detector by means of two copies of GBT-FEC protected words through two 9.6 Gbps uplinks. The board radiation tolerance is achieved by the redundancy layers in the design, using the Xilinx Soft Error Management (SEM), Triple Mode Redundancy (TMR), GBT-FEC protocol, and GBTx ASICs.

4 Radiation tests on the DB6

Four Daughterboards were separately used to test for radiation tolerance, labelled in this document as DB6-0, DB6-1, DB6-2 and DB6-3.

Two separate tests were performed to cover tolerance to Total Ionizing Dose (TID). The KU FPGA temperatures and all the currents corresponding to all the point of load regulators of the irradiated boards were monitored. Special attention was placed on the total of ten Coretek SFP+, the six KU FPGAs, the six Microsemi ProASIC3 FPGAs and the IS25LP128F reconfiguration FLASH memories. None of the components were damaged by the deposited TID: DB6-1 and DB6-2 were irradiated with Gamma radiation at ^{60}Co at the CERN CC60 facilities to 220 Gy at a rate of 3.37 Gy/h (Figure 9 of Ref.[4]) and up to 54 Gy at a rate of 0.33 Gy/h (Figure 10 of Ref. [4]), respectively. DB6-1 ran stable during the full test, however it showed an increase in the current measured in the 0.95 V power line of both sides of the board of at around 140 Gy. This was demonstrated to be correlated with the failure of the active components of the fan used to keep the KU FPGAs cool, leading to the increase of temperature on the FPGA. The currents, temperatures and functionalities of the DB6-2 were stable during the full run. DB6-3 was irradiated in the AIC-144 cyclotron proton beam line with 54 MeV protons at the IFJ in Krakow to 381 Gy for half-A FPGA and 400 Gy for half-B FPGA at a rate of 2.88 Gy/h (Figure 3). All the currents and the temperatures were stable during the full run. The higher fluctuations compared with DB6-1 (0 to 140 Gy) and DB6-2 are due to active Single Event Upset (SEU) corrections by the Xilinx SEM and resets applied when an uncorrectable SEU appeared during the run.

Two separate tests were performed for Single Event Latchups (SEL), one as described for the DB6-3 with 54 MeV protons, where the FPGAs were irradiated to a fluence of 2.62×10^{11} p/cm². In the second test, two Trezz TE0841 boards (each with the same KU FPGA used in the DB design), a Lattice ICE40LP FPGA and a ProASIC3E A3P31500 FPGA were irradiated with 226 MeV protons to a fluence of 1.11×10^{12} p/cm², for which all the currents of the FPGAs were monitored (Figure 6 of Ref. [4]). No SEL were observed in the units under test in either of the tests.

A set of tests was setup to cover the Non-Ionizing Energy Losses (NIEL) tolerance, where a DB (DB6-0) and a set of test boards with non-radiation qualified DB components were irradiated to a fluence of 14×10^{12} (1 MeV equivalent n)/cm². The post-radiation tests performed on DB6-0 demonstrated the radiation tolerance of the KU FPGAs up to the exposed fluence. In the case of the ProASIC, some after-effects could be seen where the firmware was not working as expected and re-configuration was not possible. However, the firmware functionality of all devices was recovered after annealing, with re-configuration capability recovered in some of the devices. The same applies